

Experiment No. 2

To Study the Concept of Storage as a Service (SaaS) and Implement Cloud Storage Using a Cloud Platform

Aim

To study the concept of Storage as a Service (STaaS) and implement cloud storage using GitHub as the host cloud platform.

Objectives

- To understand the core fundamentals, deployment models, and benefits of Storage as a Service.
- To configure a remote cloud repository to host and store data assets.
- To implement version control, file tracking, and public asset delivery using GitHub's cloud architecture.

Theory

Storage as a Service (STaaS)

What is Storage as a Service (STaaS)? Storage as a Service is a business model where a provider rents storage space to an individual or enterprise. It falls under the umbrella of Cloud Computing infrastructure utilities. Instead of maintaining physical hard drives or local network-attached storage (NAS), users offload hardware provisioning, redundancy, and physical security to a remote cloud data centre.

Features of STaaS

Elasticity: The ability to scale capacity upward or downward instantly based on usage spikes.

Multi-Tenancy: Multiple users or projects securely share the same underlying hardware infrastructure.

Pay-per-use: Capital expenditures (CapEx) are converted to predictable operating expenditures (OpEx)

GitHub as a STaaS Cloud Platform

While primarily a version control system, GitHub functions effectively as a public/private file and object storage service:

- **Object Mapping:** Files are tracked as blobs (binary large objects) in the cloud.
- **Data Redundancy:** Every commit creates an immutable historical version, serving as an automated backup system.
- **Content Delivery Network (CDN):** GitHub Actions and GitHub Pages allow files to be distributed via global web servers.

4. Implementation Steps (Procedural Diagram)

Follow these steps to execute the cloud storage implementation:

[Create GitHub Account] → [Create Public Repository] → [Upload Files via Browser/Git] → [Enable GitHub Pages] → [Fetch Public URL]

Step 1: Resource Provisioning

1. Log into your account at github.com.
2. Click the **New** button on the left sidebar to initialize a new storage container (Repository).
3. Set the Repository Name to `staas-cloud-storage`.
4. Set the visibility to **Public** and check **Add a README file** to generate your main storage directory.

Step 2: Data Ingestion (Uploading Assets)

1. Navigate into your newly created `staas-cloud-storage` container.
2. Click **Add file** → **Upload files**.
3. Drag and drop a test file (e.g., a PDF report or an image named `sample-image.jpg`).
4. Scroll to the bottom and click **Commit changes** to write the block of data to the cloud.

Step 3: Public Distribution Setup (GitHub Pages)

1. Open the **Settings** tab located on the repository navbar.
2. Click on **Pages** under the "Code and automation" sidebar section.
3. Under **Build and deployment**, set the Source branch to `main` (or `master`) and click **Save**.
4. Wait 60 seconds for the system to generate your dedicated cloud storage root domain.

5. Observations & Verification

After configuration, the stored data can be verified through two entry points:

File Storage Verification Table

Attribute	Observation / Value	Purpose
Storage Account Provider	GitHub Cloud Infrastructure	Remote host provider
Container/Bucket Name	staas-cloud-storage	Logical data grouping
Object File Stored	sample-image.jpg	Verified target asset
Public Retrieval URL	<code>https://<your-username>.github.io/staas-cloud-storage/sample-image.jpg</code>	Global CDN edge access point

To verify execution, copy your Public Retrieval URL, paste it into an anonymous web browser tab, and confirm that the image or file renders successfully from the remote cloud server.

6. Conclusion

By completing this practical assignment, the foundational concepts of Storage as a Service (STaaS) were successfully analysed. The experiment demonstrated that remote cloud repositories can efficiently manage, secure, track, and distribute digital objects without requiring localized hardware deployment.